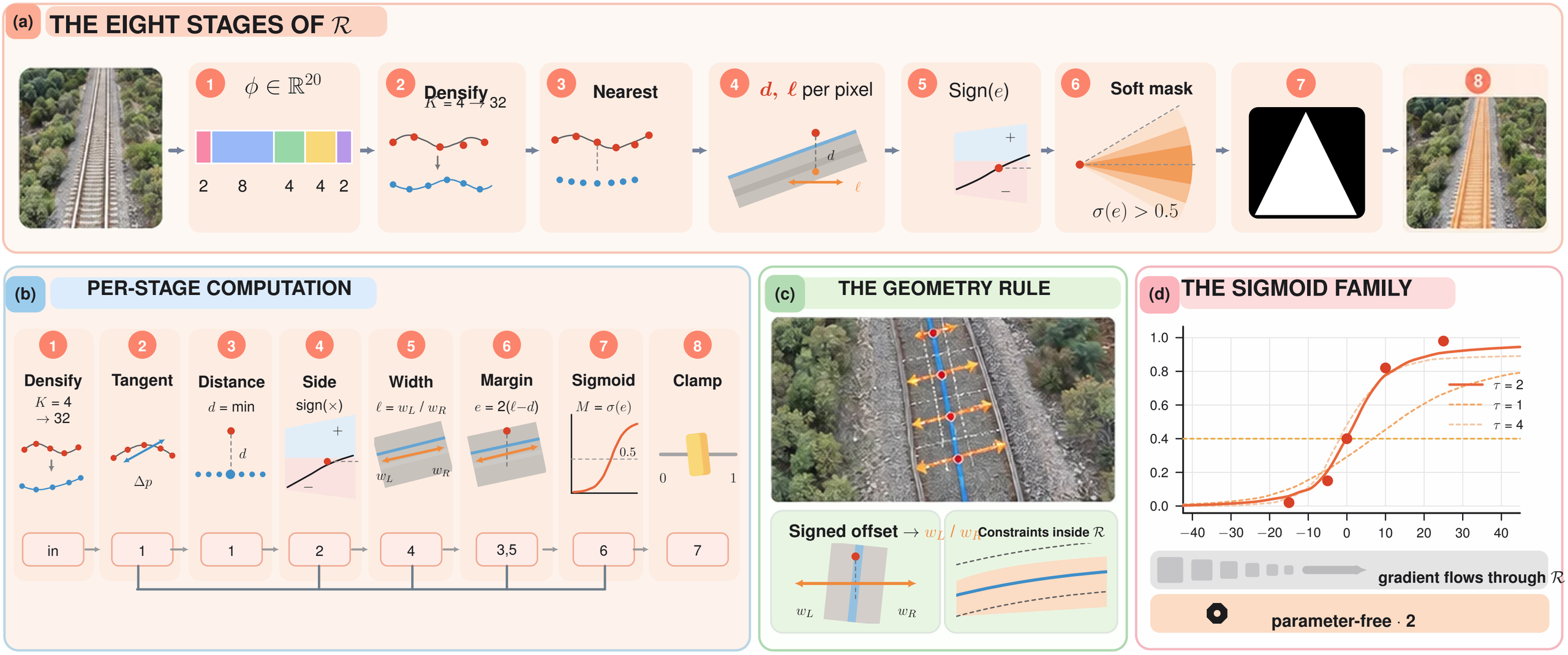

(a)
THE EIGHT STAGES OF $\mathcal{R}$
1
$\phi \in \mathbb{R}^{20}$
2 8 4 4 2
2
Densify
$K = 4 \rightarrow 32$
3
Nearest
4
d, ℓ per pixel
d
ℓ
5
Sign(e)
+
−
6
Soft mask
$\sigma(e) > 0.5$
7
8
(b)
PER-STAGE COMPUTATION
1
Densify
$K = 4$
$\rightarrow 32$
2
Tangent
Δp
3
Distance
d = min
d
4
Side
sign(×)
+
−
5
Width
$\ell = w_L / w_R$
w_L
w_R
6
Margin
$e = 2(\ell - d)$
7
Sigmoid
$M = \sigma(e)$
0.5
8
Clamp
0
1
in
1
1
2
4
3,5
6
7
(c)
THE GEOMETRY RULE
Signed offset → w_L / w_R
w_L
w_R
Constraints inside $\mathcal{R}$
(d)
THE SIGMOID FAMILY
1.0
0.8
0.6
0.4
0.2
0.0
−40 −30 −20 −10 0 10 20 30 40
$\tau = 2$
$\tau = 1$
$\tau = 4$
gradient flows through $\mathcal{R}$
parameter-free · 2